

Modeling the propagation of drying and decomposition fronts in wood

Antonio Galgano, Colomba Di Blasi *

Dipartimento di Ingegneria Chimica, Università degli Studi di Napoli "Federico II," P. le V. Tecchio, 80125 Naples, Italy

Received 10 January 2004; received in revised form 15 June 2004; accepted 20 July 2004

Available online 19 August 2004

Abstract

Decomposition of moist wood is modeled using the shrinking unreacted-core approximation for a finite rate of reaction and the assumption of a thermally controlled evaporation of moisture across an infinitely thin front at constant temperature. The one-dimensional, quasi-steady (along the char layer) equations also take into account convective, conductive, and radiative heat transfer and different physical properties for char, dry wood, and moist wood. The use of realistic values for all the input parameters results in acceptable agreement between predicted and measured weight losses for 0.04-m-thick beech wood particles (external radiative heat fluxes between 40 and 80 kW/m² and initial moisture contents between 0 and 47 wt% on a dry basis). A parametric analysis indicates that, apart from a linear dependence on the moisture content, the characteristic times are especially affected by water evaporation in the case of thick samples. It has also been found that assuming a coincident front for both wood decomposition and moisture evaporation, often done in gasification/combustion models, could be applied with sufficient accuracy only for thick samples with high moisture contents and subjected to severe thermal heating.

© 2004 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

Keywords: Modeling; Wood; Pyrolysis; Drying

1. Introduction

The importance of the thermal response of lignocellulosic materials under extreme temperature conditions has long been recognized, as they play a large part in building fires. Another application, of increasing industrial interest, is related to thermochemical conversion (pyrolysis, gasification, combustion) to recover liquid and gaseous fuels and energy. These applications have motivated the development of detailed pyrolysis models dealing with the description of all the main transport phenomena and relevant chemistry

of wood pyrolysis. The most important achievements in detailed modeling of single-particle dynamics have been reviewed by Di Blasi [1,2] and Gronli [3]. Advancements in process modeling and/or simulation have been made in more recent studies [4–16]. Transport equations are coupled with volumetric reactions for the primary formation of tars, gases, and char and, in some cases, with gas-phase cracking of tar. Such models, after further implementations concerning the heterogeneous combustion and gasification processes, could be incorporated in gasifier and combustor models. However, for such a purpose, simplifications are required to keep the computational effort within reasonable limits.

Simplified models of wood pyrolysis are usually based on the shrinking unreacted-core approxima-

* Corresponding author. Fax: 39-081-2391800.

E-mail address: diblas@unina.it (C. Di Blasi).

Nomenclature

A	Preexponential factor..... m/s	Y^*	Mass fraction of dry solid..... %
Bi	Biot number	V	Volatiles
C	Char	α	Empirical parameter (convective transport)
c	Specific heat..... J/kg K	γ	Thickness of the char layer at the beginning of the reacting stage..... m
E	Activation energy..... kJ/mol	Δh	Pyrolysis heat..... J/kg
e	Surface emissivity	η	Dry wood conversion $(1 - Y^*)/(1 - Y_c^*)$
h	Global heat transfer coefficient..... W/m ² K	ν	Stoichiometric coefficient (volatile fraction)
k^*	Thermal conductivity..... W/m K	ρ	Density..... kg/m ³
L	Particle length..... m	χ	Empirical factor
m_p	Mass flux of volatile pyrolysis products..... kg/s	ω_p	Pyrolysis rate..... kg/s
m_u	Water vapor mass flux..... kg/s	Subscripts	
m_v	Volatile mass flux per unit surface..... kg/m ² s	b	Boiling point
Nu	Nusselt number	c	Char
Pr	Prandtl number	ce	Particle center
Q	External heat flux..... W/m ²	e	External
q	Heat flux..... W/m ²	f	Final value
R	Universal gas constant..... kJ/mol K	g	Gas phase
R	Particle radius..... m	l	Liquid water
r	Radial coordinate..... m	p	Pyrolysis front
r^*	Reactivity $(-dY^*/dt)/(Y^* - Y_c^*)$ s ⁻¹	peak	Maximum value
Re	Reynolds number	s	External surface
S_p	Reaction surface..... m ²	u	Drying front
t	Time..... s	v	Volatiles
T	Temperature..... K	w	Dry virgin wood
t_c	Conversion time..... s	0	Initial
t_u	Drying time..... s	50	Dry wood conversion equal to 50%
U_0	Moisture content..... %		Parallel to the wood fibers
W	Wood	⊥	Perpendicular to the wood fibers
Y	Total condensed phase mass fraction %		

tion [17–23]. This treatment assumes that the char region is separated from the virgin wood by an infinitely thin (surface) front, where pyrolysis takes place. The solution is highly simplified by the application of integral methods to the heat-transfer problem [18,19, 21–23], owing to a change from partial to ordinary differential equations.

The most recent study [23], focused on thermochemical conversion processes, presents a comprehensive shrinking unreacted-core model for the pyrolysis of a single wood particle, taking into account unsteadiness and spatial temperature gradients for both the virgin and the charred solid, finite-rate kinetics, convective transport, and realistic property values and kinetic constants. The model has been experimentally validated and an extensive analysis of sensitivity to model assumptions and parameter values has been made. However, despite the scarce use of predried fuels in practical applications and the recognized effects

of moisture evaporation on charring rate, product yields, and gas-phase combustion of volatile products, drying processes are completely neglected, as generally done in previous shrinking unreacted-core models of wood pyrolysis. An exception is represented by the work carried out by Palchonok et al. [24], which, however, is more simplified owing to the assumptions of infinite-rate kinetics for the pyrolysis reaction, absence of convective transport, and steady state formulation of the enthalpy equations for both the char and the dried wood layers.

Moisture evaporation has been shown to increase the ignition time, the total ignition energy, and the minimum intensity for both spontaneous and piloted ignition [25]. The composition of the mixture of volatile pyrolysis products is also altered [26] with its energetic content, which, during combustion, also affects the burning rate of the solid. The increase in the initial moisture content has been reported to promote

charring reactions, to lower the liquid (oil) yields [27–29], and to reduce the maximum devolatilization rate [29]. Furthermore, a plateau in the temperature profiles, for conditions near the normal boiling point of water, has been observed [25,26,29–32]. These findings clearly indicate that the effects of moisture evaporation should be incorporated into mathematical models of wood pyrolysis, to obtain realistic predictions of process dynamics, in relation to solid flammability characteristics, and to produce reliable information for the optimal design and operation of chemical reactors.

The main aim of this study is to extend the shrinking unreacted-core model of wood pyrolysis developed by Galgano and Di Blasi [23], to include the effects of moisture evaporation, and to carry out an experimental verification using data previously published [29,33]. Furthermore, it is intended to simulate the process dynamics for widely variable conditions in order to determine the key variables and to evaluate the applicability of further simplifications in the mathematical formulation of the problem.

2. Mathematical modeling

The mathematical model considers a cylindrical wood particle (radius R_0 and length $L = 2R_0$; the cylinder axis is parallel to the wood fibers) with an initial moisture content, U_0 , referred to the initial dry wood mass, uniformly distributed throughout the particle volume. The particle is exposed to radiative heating along its lateral surface under a continuous nitrogen flow, in accordance with the experiments presented in [29,33]. The top and bottom surfaces are exposed to a relatively low temperature (about 400 K [29]), resulting from the cold (ambient) forced flow. The analysis reported in [29] evaluated the importance of internal heat conduction time versus external heat-transfer time along the radial direction, using the Biot number, $Bi_{\perp} = (eQR_0)/(k_{\perp}^* \Delta T)$, and average values for the medium properties ($e = 0.8$, $k_{\perp}^* = 0.14$ W/(m K), and the temperature difference ΔT referred to the pyrolysis temperature). For external heat fluxes in the range 40–80 kW/m², which are of interest in this study, the Biot number was found to vary between 13 and 25, indicating a controlling role of internal heat transfer.

As for the top and bottom surfaces, exposed to convective heating only, the Biot number can be defined as $Bi_{\parallel} = hL/2k_{\parallel}^*$, with a global heat-transfer coefficient evaluated by the empirical correlation $Nu = 0.332Pr^{1/3}Re^{1/2}$, valid for a flat plate exposed to a parallel laminar flow [34], with a velocity of 0.0125 [29], and $k_{\parallel}^* = 0.28$ W/(m K) (the thermal conductivity parallel to the wood fibers is roughly

two times higher than that along the perpendicular direction [35]). The figures, obtained in this case for external temperatures in the range 400–600 K, vary between 0.19 and 0.22, indicating that the external heat transfer rate is controlling. Therefore, given values below 1, the temperature variation along the cylinder axis can be ignored and the mathematical description, as already in [23], uses the assumption of one-dimensional process along the particle radius.

Wood decomposition and transport phenomena are described using the unreacted-shrinking-core approximation and have already been discussed in detail in [23]. In this study moisture evaporation is also taken into account and described, assuming that it takes place at an infinitely thin, constant-temperature front (T_b), which separates the moist from the dry region. Furthermore, it is also assumed that the evaporation temperature at the drying front coincides with the normal boiling point of water; the moist core of the particle is at ambient (initial) temperature; and the heat flux at the drying front is applied exclusively to raising the surface temperature, from the initial to the evaporation value, and sustaining the endothermic evaporation process (λ enthalpy of evaporation).

During moisture evaporation and wood decomposition, the particle, initially at ambient conditions and suddenly exposed to thermal radiation, undergoes several stages. The model assumes that drying begins at time $t = 0$, as soon as the external radiative heat flux is turned on. During this stage (initial drying) the activity of the decomposition reaction is negligible and the size of the already dried wood extends from the drying front ($r = R_u$) to the external surface ($r = R_0$). This is followed by the simultaneous presence of both wood decomposition and moisture evaporation (decomposition of moist wood). The passage from the first to the second stage is assumed to occur when the thickness of the char layer equals a small assigned fraction of the wood particle radius, γ . As the decomposition reaction becomes active, the integration domain comprises three zones: the char layer ($R_p \leq r \leq R_0$), the dry wood ($R_u \leq r \leq R_p$), and the moist core ($0 \leq r \leq R_u$) (Fig. 1 provides a schematization of the problem). Finally, when drying is completed, only two zones remain: the char layer ($R_p \leq r \leq R_0$) and the unreacted dry wood ($0 \leq r \leq R_p$) (decomposition of dry wood).

The model equations are listed in Appendix A for the decomposition of moist wood (char, unreacted dry wood, and moist wood zones) and specifications are also reported for the initial drying stage and the decomposition of dry wood. The solution is obtained by means of the heat balance integral method and the computational procedure is the same as already described in detail in [23] for the pyrolysis of dry wood.

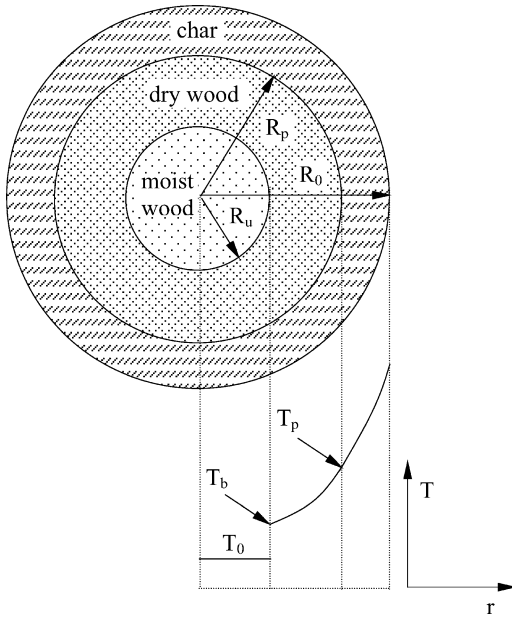


Fig. 1. Schematic representation of the moist wood particle undergoing pyrolysis (cross section of the particle).

The properties of the virgin solid and char and the kinetic constants of the pyrolysis reaction are representative of beech wood. The values, reported in Table 1, are those already used in [23] and derived, in large part, from previous literature. Moisture properties are also derived from previous literature. In particular, the enthalpy of water evaporation is assumed to depend on the moisture content according to the relation [36]

$$\lambda = 3.348 - 13.085U + 60.262U^2 - 95.778U^3, \quad U \leq 0.3, \quad (1)$$

$$\lambda = 2.260, \quad U > 0.3,$$

where the variable λ is expressed as kJ/g. To take into account the variation in the moisture content, at the evaporation front, a mean value is used. It is computed by integration of Eq. (1).

$$\tilde{\lambda} = \frac{\int_0^{U_0} \lambda dU}{U}, \quad (2a)$$

giving

$$\tilde{\lambda} = 3.348 - 6.542U_0 + 20.087U_0^2 - 23.944U_0^3, \quad U_0 \leq 0.3, \quad (2b)$$

$$\tilde{\lambda} = \frac{2.547 + 2.260(U_0 - 0.3)}{U_0}, \quad U_0 > 0.3.$$

The heating conditions used in the simulations try to reproduce those established in the experiments [29], where beech wood cylinders (with $R_0 =$

Table 1

Property values and kinetic data of beech wood pyrolysis (reference data)

Parameter	Value	Ref.
A	4.37×10^6 (m/s)	[23]
c_v	1500 (J/kg K)	[3]
c_u	2100 (J/kg K)	[40]
c_l	4200 (J/kg K)	[40]
c_w	1500 (J/kg K)	[23]
e_c	1.0	[22]
e_w	0.6	[22]
E	143 (kJ/mol)	[41]
k_c^*	0.42 (W/m K)	[23]
k_w^*	0.3 (W/m K)	[23]
α	0.75	[23]
Γ	$10^{-4} R_0$ (m)	[23]
Δh	4.3×10^5 (J/kg)	[23]
χ	0.7	[23]

Note. $R_0 = 0.02$ m, $L = 0.04$ m, $U_0 = 0.27$, $Q_f = 49$ kW/m², $v_v = 0.751$, $T_e = 400$ K, $\rho_w = 700$ kg/m³ [29].

0.02 m and $L = 0.04$ m) were vertically positioned in the uniformly heated zone of a radiant furnace, by means of a quartz reactor. Because of the assumption of infinitely thin evaporation and decomposition fronts, only the range of high radiation intensities, 40–80 kW/m², are considered. In accordance with the simulations carried out for dry wood [23], a time of 300 s is assumed to correspond to the attainment of the final value of the heat flux, Q_f . The time dependence of the heat flux, Q , is linear and comprises a fast and a slow stage. In the former, it attains 95% of its final value, with a duration of about 200–50 s which can be expressed by the following empirical relation:

$$t_{\text{int}} = 50.9 + 17,922 \exp(-0.119Q_f). \quad (3)$$

The absorption of radiation from the volatile pyrolysis products and steam and the reduced transparency of the quartz reactor due to tar deposition during wood degradation are taken into account by the empirical parameter χ . The temperature of the bulk gas phase adjacent the sample, T_e , was estimated to be 400 K [29].

3. Results

The dynamics of particle drying and decomposition are discussed first for the reference case, using the data listed in Table 1. Then a parametric study, concerning the pyrolysis of moist ($U_0 = 0$ –100%) particles with radii comprised between 0.01 and 0.1 m and $Q_f = 40$ –100 kW/m², and a sensitivity analysis to input parameters are carried out. Finally, an extensive comparison between model predictions and ex-

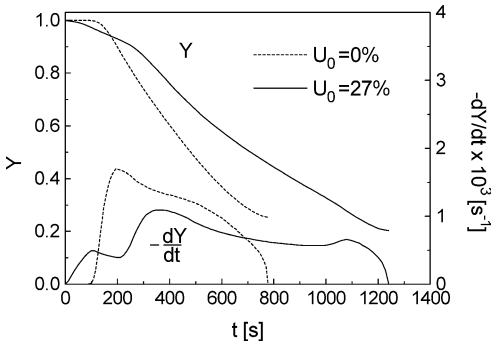


Fig. 2. Solid mass fraction, Y , and devolatilization rate, $-dY/dt$, as functions of time as predicted for $U_0 = 0$ (dotted lines) and $U_0 = 27\%$ (solid lines) (reference data, Table 1).

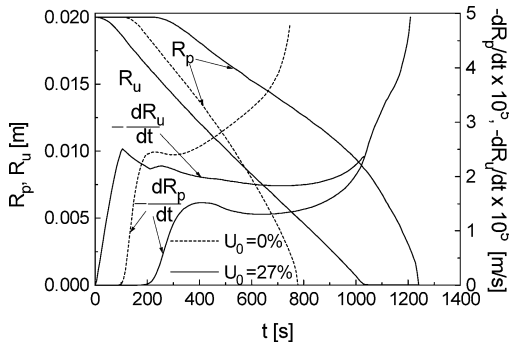


Fig. 3. Positions R_u , R_p and propagation rates $-dR_u/dt$, $-dR_p/dt$, of the evaporation and pyrolysis fronts, respectively, as functions of time as predicted for $U_0 = 0$ (dotted lines) and $U_0 = 27\%$ (solid lines) (reference data, Table 1).

perimental measurements is provided by varying U_0 in the range 0–47% ($Q_f = 49 \text{ kW/m}^2$) and Q_f in the range 40–80 kW/m^2 ($U_0 = 11\%$).

3.1. Process dynamics

Predictions of the process dynamics for the reference data (Table 1) in the case of moist ($U_0 = 27\%$) and dry wood can be seen from Fig. 2 (solid mass fraction and devolatilization rate as functions of time), Fig. 3 (position and propagation rate of the evaporation and pyrolysis fronts as functions of time), Fig. 4 (reactivity, $r^* = (-dY^*/dt)/(Y^* - Y_C^*)$, as a function of the dry wood conversion, $\eta = (1 - Y^*)/(1 - Y_C^*)$), and Fig. 5 (temperatures at the surface, pyrolysis front and center of the particle as functions of time).

At first glance it appears that the presence of moisture results in longer conversion times, slower devolatilization rates and propagation speeds of the decomposition front, and, on the average, lower tem-

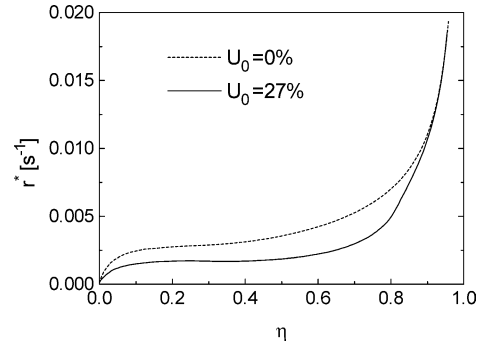


Fig. 4. Reactivity, $r^* = (-dY^*/dt)/(Y^* - Y_C^*)$, as a function of dry wood conversion, $\eta = (1 - Y^*)/(1 - Y_C^*)$, as predicted for $U_0 = 0$ (dotted line) and $U_0 = 27\%$ (solid line) (reference data, Table 1).

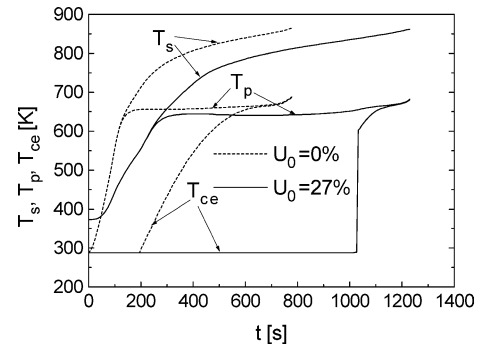


Fig. 5. Temperatures at the surface, T_s , pyrolysis front, T_p , and center, T_{ce} , of the particle as functions of time as predicted for $U_0 = 0$ (dotted line) and $U_0 = 27\%$ (solid line) (reference data, Table 1).

peratures. Some qualitative features of dry wood decomposition are preserved but significant changes are observed for very short and very long times.

An initial delay is observed for dry wood, following a preheating period to achieve decomposition temperatures, whereas in the presence of moisture, mass loss begins at time $t = 0$ (Fig. 2). Indeed, when the computational procedure is started, the external heat flux is entirely applied for surface heating and moisture evaporation. Consequently, an evaporation front also appears immediately (Fig. 3). The rate of moisture release and the propagation speed of the evaporation front initially increase linearly with time. Then, as soon as the thickness of the dried wood becomes nonnegligible (a time of about 100 s), they start to decrease and attain a local minimum (a time of about 200 s). At this time, the decomposition process also begins. In this way, the large amount of volatiles generated from wood decomposition adds to steam and gives rise to a new increase in the devolatilization rate. The maximum (at about 400 s) is significantly

lower and more delayed than that predicted for dry wood.

The propagation rate of the decomposition front shows the same qualitative dependence on time (Fig. 3) independent of the moisture content. Moreover, in this case, the evaporation and decomposition fronts propagate simultaneously for a large part of the process. Apart from the initial and final transients, both fronts move with almost constant speed (especially the evaporation front). The rapid increase in the propagation speeds, as complete conversion is approached, is due to the reduction in the size of the wet core and the increase in the pyrolysis temperature for both moist and dry samples. Following a wide zone of decreasing values, ensuing from variable temperature and reaction surface at the decomposition front, the devolatilization rate of the moist particle presents a second local peak, associated with the completion of the endothermic evaporation process and the increase in the amount of heat available to preheat and decompose the already dried wood. Given the integral solution method, the completion of the evaporation process is also associated with a sudden change in the shape of the temperature profile, as the boundary condition at the inner side changes from an assigned (evaporation) temperature to a zero heat flux. This feature is completely absent for the case of dry wood. The conversion time is augmented by about 67% as a consequence of $U_0 = 27\%$.

To quantify the influences of moisture evaporation on the decomposition rate of wood, the reactivity is reported as function of the conversion of dry wood in Fig. 4 for the moist and dry particles. An initial moisture content of 27% causes a reduction in the reactivity of wood by a factor of about 50% for conversions in the range 0–95%. Only for higher conversions and as a consequence of the local peak in the devolatilization rate simulated for $t = 1100$ s (Fig. 2) and pyrolysis temperature of 1–2 K higher, the moist sample becomes barely more reactive.

The time evolution of the temperature at the particle surface and the pyrolysis front are qualitatively similar for both dry and moist wood (Fig. 5) and reproduce the same trends already discussed [23]. In particular, for the moist particle, the pyrolysis temperature rapidly increases and remains roughly constant as long as moisture evaporation is active. The conclusion of the endothermic evaporation process and the symmetry condition at the particle center are responsible, over long times, for the increasing T_p values. The temperature at the particle center, T_{ce} , is also constant at its initial value while moisture evaporation is underway. Then, it rapidly attains the pyrolysis value.

The temperature of the moist core remains at the initial value. An alternative treatment, still avoiding the description of the heat transfer across the moist

core, could foresee a preheating stage to bring the temperature of the entire region to the boiling point of water, before evaporation begins along the external surface. However, the unavoidable delay in the mass loss, as shown in the following, is contradicted by measurements.

3.2. Influences of the initial moisture content

To investigate the role of the initial moisture content on the degradation characteristics, a parametric analysis, consisting of three sets of simulations, has been carried out. The initial moisture content has been varied in the range 0–100% ($R_0 = 0.02$ m and $Q_f = 49$ kW/m²). Then simulations have been made for $U_0 = 0$ and $U_0 = 50\%$ by varying either the particle radius between 0.01 and 0.1 m ($Q_f = 49$ kW/m²) or the external radiative heat flux between 40–100 kW/m² ($R_0 = 0.02$ m). Simulation results are shown in terms of drying time (95% evaporation of moisture), t_u , conversion time (95% conversion of dry wood), t_c , the temperature of the decomposition front for 50% conversion of dry wood, T_{p50} , reactivity corresponding to 50% conversion of dry wood, r_{50}^* , and maximum devolatilization rate, $-dY/dt_{peak}$.

The drying and conversion times present a linear dependence on the initial moisture content (Fig. 6A), as reported by previous experimental and theoretical analysis [29,37,38]. The pyrolysis temperature, T_{p50} , the reactivity, r_{50}^* , and the peak rate, $-dY/dt_{peak}$, are successively reduced as the initial moisture content is increased (Figs. 6A and 6B), owing to the successively higher amounts of heat required at the evaporation front. The variations are relatively small for the temperature (from 649 to 623 K), whereas the peak rate and the reactivity are reduced approximately by factors of 2 and 3, respectively, over the range of U_0 examined.

The characteristic times become successively longer as the particles become thicker. The qualitative dependence is not altered by moisture evaporation though, as expected, they are quantitatively different (Fig. 7A). Moisture evaporation effects are comparatively stronger as successively thicker particles are examined. For instance, for a particle with $U_0 = 50\%$ the conversion time increases by a factor of 98% for $R_0 = 0.01$ m and a factor of 155% for $R_0 = 0.1$ m with respect to the case of dry samples. The temperature, T_{p50} , decreases as the particle radius becomes larger (Fig. 7A), as a consequence of the enhanced heat transfer resistance. From the quantitative point of view, the T_p dependence on R_0 is initially rather strong with variations comprised between 665 and 617 K as R_0 varies from 0.01 to 0.03 m in the case of dry wood. On the other hand, variations caused

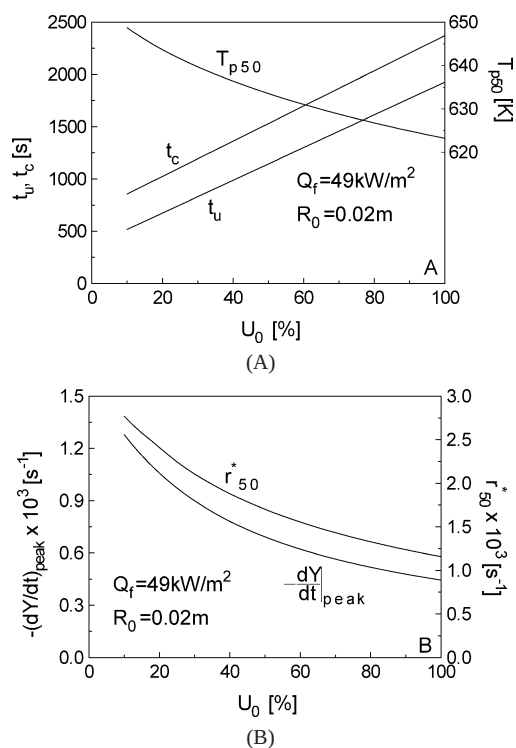


Fig. 6. (A) Predictions of the drying time, t_u , the conversion time, t_c , and the temperature of the decomposition front for 50% conversion of dry wood, T_{p50} , as functions of the initial moisture content, U_0 (reference data, Table 1); (B) predictions of the reactivity corresponding to 50% conversion of dry wood, r_{50}^* , and maximum devolatilization rate, $-(dY/dt)_{\text{peak}}$, as functions of the initial moisture content, U_0 (reference data, Table 1).

by moisture evaporation are relatively small (about 7–14 K) over the range of R_0 considered. However, given the higher values, the percent variation is smaller at low radii. The peak devolatilization rate and reactivity also become successively lower as R_0 is increased, as a consequence of the enhanced internal heat transfer resistance. In quantitative terms, the maximum peak rate is roughly halved as the moisture level is increased from 0 to 50%, independently of the particle size. On the contrary, its effect on the reactivity is slightly stronger for small radii (reductions by factors of 1.9 for $R_0 = 0.01 \text{ m}$ and 1.7 for $R_0 = 0.1 \text{ m}$).

As expected, characteristic times become successively shorter (Fig. 8A) as the external heat flux is increased. The effects of moisture evaporation are more or less the same (percent variation of 124 and 130% for $Q_f = 40$ and 100 kW/m^2 , respectively, for an increase in U_0 from 0 to 50%). The dependence of the pyrolysis temperature on Q_f is linear and an initial moisture content of 50% introduces a diminution of about 20 K independently of the heating conditions

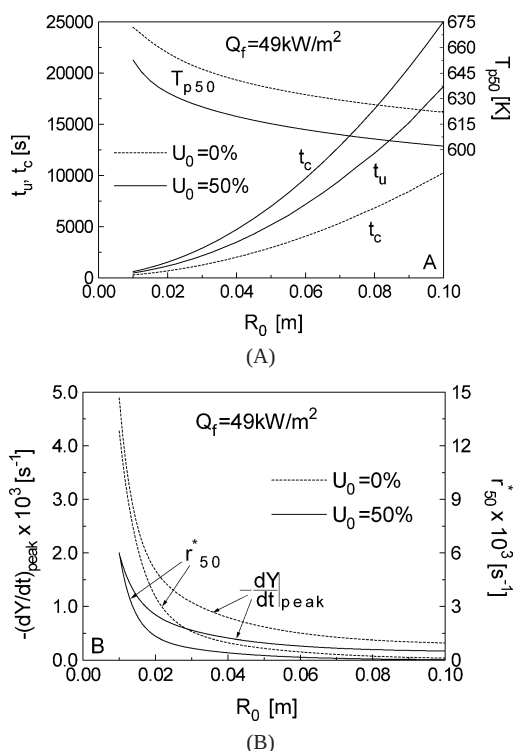


Fig. 7. (A) Predictions of the drying time, t_u , the conversion time, t_c , and the temperature of the decomposition front for 50% conversion of dry wood, T_{p50} , as functions of the initial particle radius, R_0 (reference data, Table 1); (B) predictions of the reactivity corresponding to 50% conversion of dry wood, r_{50}^* , and maximum devolatilization rate, $-(dY/dt)_{\text{peak}}$, as functions of the initial particle radius, R_0 (reference data, Table 1).

(Fig. 8A). The peak rate and reactivity increase with Q_f and are more than halved for $U_0 = 50\%$ compared with the case of dry wood (Fig. 8B). In other words, the effects of moisture evaporation are percentage-wise about the same as Q_f is varied.

Information on the process dynamics as the moisture content and the particle size are varied can be obtained from Fig. 9A, where the thickness of the dry wood layer is plotted as a function of time ($Q_f = 49 \text{ kW/m}^2$). In all cases, it initially increases to a maximum, as a consequence of the faster propagation speed of the evaporation front with respect to the decomposition front. Then, as moisture evaporation terminates and the decomposition front moves toward the particle center, it rapidly goes to zero. The dry wood layer also becomes successively thinner as R_0 or U_0 is increased; that is, moisture evaporation becomes more strongly coupled with wood decomposition. For the strongest coupling ($U_0 = 50\%$ and $R_0 = 0.1 \text{ m}$), the dry wood layer only varies between about 5 and 20% of R_0 . This finding supports the assumption, made in some previous studies, of wood

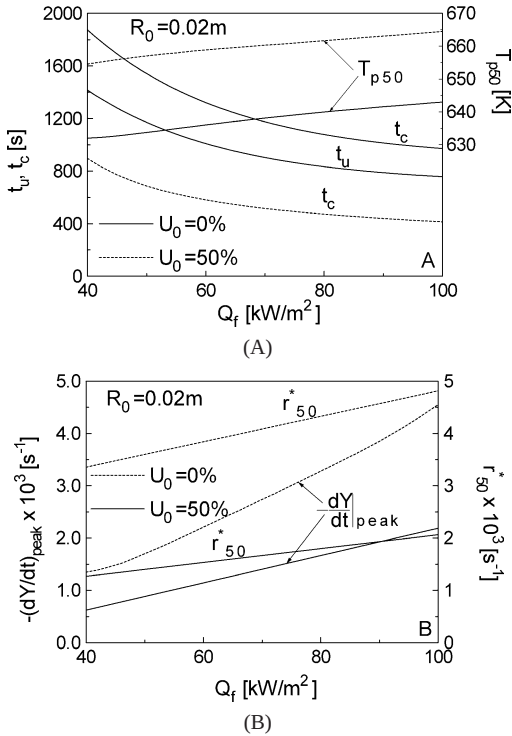


Fig. 8. (A) Predictions of the drying time, t_u , the conversion time, t_c , and the temperature of the decomposition front for 50% conversion of dry wood, T_{p50} , as functions of external heat flux, Q_f (reference data, Table 1); (B) predictions of the reactivity corresponding to 50% conversion of dry wood, r_{50}^* , and maximum devolatilization rate, $-dY/dt_{peak}$, as functions of the external heat flux, Q_f (reference data, Table 1).

gasification/combustion [39] of a single infinitely thin front where both moisture evaporation and wood decomposition take place. On the other hand, this simplification is clearly not applicable for thin particles (for instance, the case of $R_0 = 0.02$ m), where the maximum thickness of the dry wood layer may attain values of about 50% of the initial particle radius.

For thick particles ($R_0 = 0.1$ m), the assumption of a single evaporation/decomposition front becomes progressively less accurate as the heating conditions are made less severe, as clearly shown in Fig. 9B (the dry wood layer as a function of time for several heat fluxes and two moisture levels).

3.3. Sensitivity analysis

An analysis of sensitivity of the model predictions to variations in the input parameters has been carried out, following the same methodology already presented by Galgano and Di Blasi [23]. A sensitivity parameter is introduced, defined as the ratio between the % of the deviation of outputs (the +

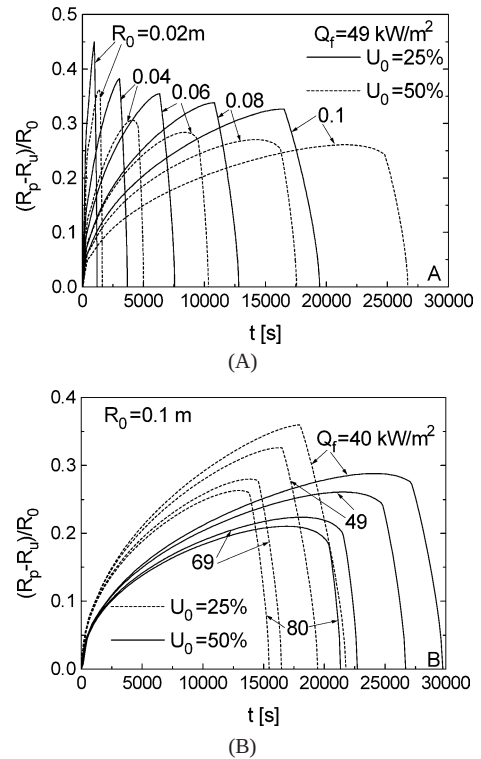


Fig. 9. (A) Predictions of the dry wood layer, expressed as a fraction of the initial radius, as a function of time for different particle radius and two initial moisture contents (reference data, Table 1); (B) predictions of the dry wood layer, expressed as a fraction of the initial radius, as a function of time for different external heat fluxes and two initial moisture contents (reference data, Table 1).

and – signs indicate an increase and decrease from the reference value, respectively) to the % deviation of inputs (again with respect to the reference value). Deviations of $\pm 25\%$ have been considered for the input parameters (variations on T_b refer to $^{\circ}\text{C}$). Reference values of the output variables are those predicted by means of the input data listed in Table 1. Output variables examined include the drying (t_u) and the conversion (t_c) times, the temperature at the pyrolysis front for 50% of dry wood conversion, T_{p50} , the peak rate, $-dY/dt_{peak}$, and the reactivity corresponding to 50% of dry wood conversion, r_{50}^* . Results, reported in Table 2 only for values of the sensitivity parameter ≥ 0.2 , confirm the findings previously discussed for dry wood [23] (highest sensitivities for variations in the variables E , Q_w , χ , and e). As for the parameters specific to moist wood, the strongest effects are caused by the initial moisture content and the vaporization enthalpy on all the output variables, except for the temperature at the pyrolysis front, whose variations are comparatively smaller.

Table 2
Sensitivity of model outputs to input parameters

Sensitivity	t_u	t_c	T_{p50}	$-dY/dt_{peak}$	r_{50}^*
c_w^+	0.26	0.27	-0.01	-0.26	-0.24
c_w^-	0.28	0.27	-0.01	-0.31	-0.32
E^+	-0.08	2.56	0.99	-1.99	-0.11
E^-	-0.03	0.84	0.95	-0.97	-2.16
e_c^-	-0.22	-0.31	0	0.86	0.05
e_w^+	-0.22	-0.14	0	-0.02	-0.03
e_w^-	-0.39	-0.28	0	-0.11	-0.06
k_c^{*+}	-0.09	-0.27	0.02	0.15	0.59
k_c^{*-}	-0.11	-0.43	0.02	0.19	0.61
k_w^{*+}	-0.38	-0.18	0.02	0.02	0.43
k_w^{*-}	-0.47	-0.26	0.01	-0.01	0.32
T_e^+	-0.07	-0.13	0.01	0.36	0.17
Δh^+	0.03	0.17	0	-0.20	-0.14
Δh^-	0.07	0.17	-0.01	-0.23	-0.15
ρ_w^+	0.94	0.92	-0.03	-0.77	-0.76
ρ_w^-	0.96	0.93	-0.04	-1.20	-1.28
χ^+	-0.55	-0.62	0.02	1.40	0.50
χ^-	-0.94	-1.20	0.01	1.41	0.29
U_0^+	0.54	0.39	-0.01	-0.32	-0.38
U_0^-	0.59	0.40	-0.02	-0.39	-0.49
T_b^+	0.26	0.14	-0.01	-0.04	-0.17
T_b^-	0.29	0.12	-0.01	-0.04	-0.20
λ^+	0.48	0.34	-0.01	-0.29	-0.32
λ^-	0.52	0.34	-0.01	-0.34	-0.40

Note. Sensitivity is the ratio between the % of the deviation of outputs, with respect to the reference values, to $\pm 25\%$ deviation of inputs, again with respect to reference values reported in Table 1; variations in T_b refer to $^{\circ}\text{C}$. Only sensitivities greater than 0.2 are listed.

3.4. Model validation

In order to validate the model, measurements [29,33] for the weight loss characteristics of beech wood cylinders with $R_0 = 0.02\text{ m}$ and $L = 0.04\text{ m}$ exposed to radiative heating have been considered. Simulations have been carried out by means of the reference data listed in Table 1 and Eq. (3) for the variable t_{int} . Furthermore, as in the simulations of dry wood particles, the stoichiometric coefficients for the total volatile yields are those measured in the experiments. In particular, the following values have been used: $\nu_v = 0.745\text{--}0.768$ for $Q_f = 49\text{ kW/m}^2$ and $U_0 = 0\text{--}47\%$, $\nu_v = 0.74\text{--}0.79$ for $U_0 = 11\%$ and $Q_f = 40\text{--}80\text{ kW/m}^2$.

Predictions and measurements, obtained for different U_0 and $Q_f = 49\text{ kW/m}^2$, are compared for the solid mass fractions and the devolatilization rate versus time in Figs. 10 and 11, respectively. The model is capable of predicting the main qualitative features of the weight loss curves apart from very short times. Indeed, it appears that the presence of a minimum in the global devolatilization rate, absent from the

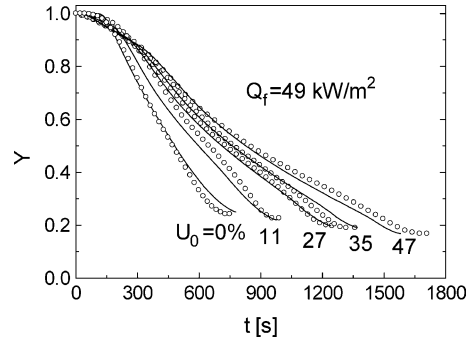


Fig. 10. Solid mass fraction, Y (solid lines), as functions of time as predicted (reference data, Table 1) for several initial moisture contents. Experimental measurements (symbols) derived from Di Blasi et al. [29,33] are also reported.

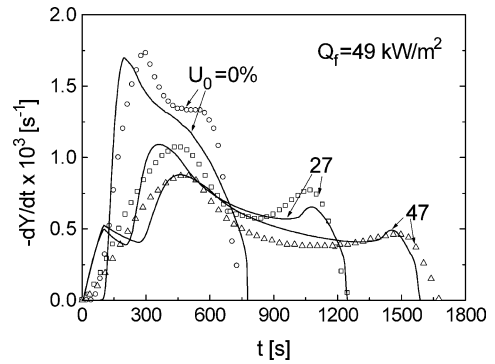


Fig. 11. Devolatilization rate, $-dY/dt$ (solid lines), as functions of time as predicted (reference data, Table 1) for several initial moisture contents. Experimental measurements (symbols) derived from Di Blasi et al. [29,33] are also reported.

measured curves, is due to model assumptions. As expected, the process becomes successively slower with reduced peak rates as the initial moisture content is increased. Indeed, both the amount of water to be heated and evaporated and the heat convected out from the particle become successively higher. For the range of moisture contents examined, moisture evaporation and wood pyrolysis always occur simultaneously, though at different positions along the particle radius. From the quantitative point of view the agreement between predictions and measurements is acceptable.

A further comparison between predictions and measurements is made through Figs. 12 and 13, obtained for an initial moisture content of 11% and several heat fluxes. Again, apart from the prediction of a barely visible minimum in the devolatilization rate at short times, the model reproduces correctly the qualitative trends. Furthermore, predictions are also quantitatively acceptable.

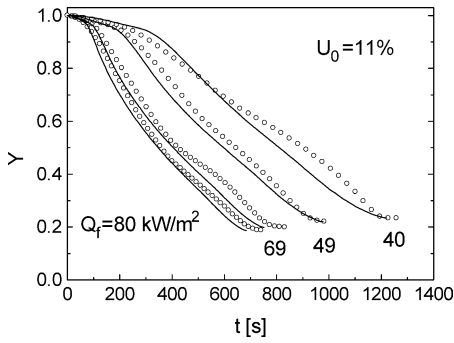


Fig. 12. Solid mass fraction, Y (solid lines), as functions of time as predicted (reference data, Table 1) for several external heat fluxes and $U_0 = 11\%$. Experimental measurements (symbols) derived from Di Blasi et al. [29] are also reported.

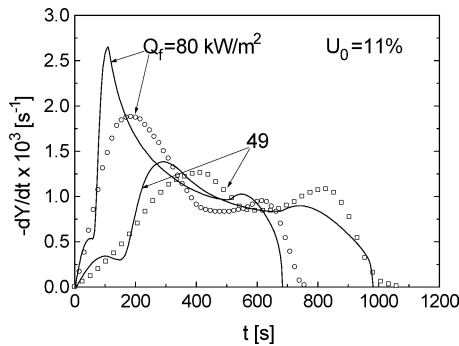


Fig. 13. Devolatilization rate, $-dY/dt$ (solid lines), as functions of time as predicted (reference data, Table 1) for several external heat fluxes and $U_0 = 11\%$. Experimental measurements (symbols) derived from Di Blasi et al. [29] are also reported.

4. Conclusions

The shrinking unreacted-core model of wood pyrolysis, previously developed and validated [23], has been extended to include the effects of moisture evaporation. This process, which is of great importance in practical applications and is known to affect the process variables dramatically, was not taken into account in previous simplified models of wood pyrolysis. In this respect, the effort done in this study is a significant advance in the development of reliable single-particle models to be coupled with CFD models for reactor simulation.

Moisture evaporation has been assumed to occur at an infinitely thin evaporation front at a known temperature coincident with the normal boiling point of water. The model correctly predicts the qualitative trends observed in the experiments of single particles exposed to thermal radiation. Furthermore, a comparison with measured weight loss characteristics, obtained for a wide range of initial moisture contents

and external heat fluxes, indicate that simulation results are also acceptable from the quantitative point of view.

As expected the drying and conversion times present an almost linear dependence on the initial moisture content. Moreover, though the effects of this parameter do not appear to depend on the external heating conditions, they become quantitatively more important for successively thicker particles.

The initial moisture content also highly affects other process variables (reactivity, pyrolysis temperature, maximum devolatilization rate), but percent variations are practically independent of both particle size and external heat flux.

The model has also been used to assess the accuracy of the assumption, made in previous studies of wood gasification/combustion, concerning the existence of a single front where moisture evaporation and wood decomposition occur simultaneously. It has been found that this treatment could be applied, though with some approximation, only for very thick and moist samples under high level heat fluxes.

Appendix A

A.1. Decomposition of moist wood—char layer ($R_p \leq r \leq R_0$)

For thick particles, exposed to high radiative heat fluxes, wood decomposition begins well before moisture evaporation is completed. Hence, the simultaneous propagation of a drying and a decomposition front should be modeled. When $R_p \neq R_0$, the steady enthalpy equation for the char layer, with the boundary conditions, is expressed as

$$0 = -\alpha \frac{1}{r} \frac{\partial}{\partial r} [r m_v c_g T] + \frac{1}{r} \frac{\partial}{\partial r} \left[k_c^* r \frac{\partial T}{\partial r} \right], \quad (\text{A.1a})$$

$$k_c^* \frac{\partial T}{\partial r} \Big|_{r=R_0} = q_s, \quad (\text{A.2a})$$

$$T \Big|_{R_p^+} = T \Big|_{R_p^-}. \quad (\text{A.2b})$$

The continuity equation for the gas phase and the related boundary condition at the pyrolysis front are formulated as

$$\frac{1}{r} \frac{d}{dr} (r m_v) = 0, \quad (\text{A.3a})$$

$$S_p m_v \Big|_{R_p^+} = m_u + m_p, \quad (\text{A.3b})$$

giving

$$r m_v = \text{const} = \frac{m_u + m_p}{2\pi L}. \quad (\text{A.3c})$$

Hence, the enthalpy conservation can be rewritten as

$$0 = -\alpha \frac{m_u c_u + m_p c_v}{2\pi r L} \frac{\partial T}{\partial r} + \frac{1}{r} \frac{\partial}{\partial r} \left[k_c^* r \frac{\partial T}{\partial r} \right], \quad (\text{A.1b})$$

where α is introduced to take into account that fractions of volatile pyrolysis products and steam are expelled through char cracks before thermal equilibrium is achieved.

A.2. Decomposition of moist wood—dry wood layer ($R_u \leq r \leq R_p$)

In the presence of the decomposition reaction along the dry wood zone, the mass conservation for the solid is

$$\frac{dR_p}{dt} = -A \exp\left(-\frac{E}{RT_p}\right), \quad (\text{A.4a})$$

$$R_p(0) = R_0, \quad (\text{A.4b})$$

whereas the enthalpy conservation is

$$\rho_w c_w \frac{\partial T}{\partial t} = -\alpha \frac{1}{r} \frac{\partial}{\partial r} [r m_v c_u T] + \frac{1}{r} \frac{\partial}{\partial r} \left[k_w^* r \frac{\partial T}{\partial r} \right]. \quad (\text{A.5a})$$

Equation (A.5a) is associated with the boundary conditions

$$S_p q_c = S_p q_w + m_p \Delta h \quad (r = R_p), \quad (\text{A.6a})$$

$$T|_{R_u} = T_b \quad (r = R_u), \quad (\text{A.6b})$$

where

$$q_c = k_c^* \frac{\partial T}{\partial r} \Big|_{R_p^+}, \quad (\text{A.7a})$$

$$q_w = k_w^* \frac{\partial T}{\partial r} \Big|_{R_p^-}. \quad (\text{A.7b})$$

A.3. Decomposition of moist wood—moist wood layer ($0 \leq r \leq R_u$)

In accordance with the assumptions listed above, the balance of the heat fluxes at the drying front is formulated as

$$S_u q_u = m_u [\tilde{\lambda} + c_l (T_b - T_0)] + \frac{m_u}{U_0} c_w (T_b - T_0), \quad (\text{A.8})$$

where

$$S_u = 2\pi L R_u, \quad (\text{A.9a})$$

$$q_u = k_w^* \frac{\partial T}{\partial r} \Big|_{r=R_u}. \quad (\text{A.9b})$$

In accordance with the model assumptions, the total heat flux reaching the front, $S_u q_u$, is used to raise the temperature of both moisture and dry wood (from the

initial ambient value to the water boiling point) and to evaporate moisture.

The conservation equation of the condensed-phase mass provides the position of the drying front as a function of time:

$$\frac{dR_u}{dt} = -\frac{m_u}{2\pi R_u L \rho_w U_0}, \quad (\text{A.10a})$$

$$R_u(0) = R_0. \quad (\text{A.10b})$$

A.4. Initial drying—dry wood layer ($R_u \leq r \leq R_p$)

As soon as $R_u \neq R_0$, the enthalpy conservation equation (A.5a) for dry wood is considered. The boundary condition (Eq. (A.6b)) still holds, but Eq. (A.6a) is replaced by the equation

$$k_w^* \frac{\partial T}{\partial r} \Big|_{R_0} = q_s. \quad (\text{A.6c})$$

The gas-phase continuity equation and the associated boundary condition at the drying front, respectively, are

$$\frac{1}{r} \frac{d}{dr} (r m_v) = 0, \quad (\text{A.11a})$$

$$S_u m_v \Big|_{R_u^+} = m_u, \quad (\text{A.11b})$$

giving

$$r m_v = \text{const} = \frac{m_u}{2\pi L}. \quad (\text{A.11c})$$

By combining Eqs. (A.5a) and (A.11c), the enthalpy conservation equation becomes

$$\rho_w c_w \frac{\partial T}{\partial t} = -\alpha \frac{m_u c_u}{2\pi r L} \frac{\partial T}{\partial r} + \frac{1}{r} \frac{\partial}{\partial r} \left[k_w^* r \frac{\partial T}{\partial r} \right]. \quad (\text{A.5b})$$

In order to evaluate the critical thickness of the char layer, γ , for starting the simulation of the decomposition process, the mass conservation equation for dry wood (Eq. (A.4a)) is also solved during this stage.

A.5. Initial drying—moist wood core ($0 \leq r \leq R_u$)

The conservation of enthalpy and mass is expressed by Eqs. (A.8)–(A.10). However, for the time $t = 0$, Eqs. (A.11a) and (A.11b) are replaced by the equations

$$S_u = 2\pi L R_0, \quad (\text{A.9c})$$

$$q_u = q_s. \quad (\text{A.9d})$$

A.6. Dry wood decomposition—dry wood and char layers

The enthalpy conservation for dry wood is obtained from Eq. (A.5b) where $m_u = 0$. The boundary condition (Eq. (A.6a)) still applies, but Eq. (A.6b) is

replaced by the symmetry condition,

$$k_w^* \frac{\partial T}{\partial r} \Big|_{r=0} = 0. \quad (\text{A.6d})$$

The enthalpy (Eq. (A.1a)) and mass (Eq. (A.3a)) conservation equations for the char layer are associated with boundary conditions (Eqs. (A.2a) and (A.2b)) and (Eq. (A.3b)), respectively, where $m_u = 0$.

References

- [1] C. Di Blasi, Prog. Energy Combust. Sci. 19 (1993) 71–104.
- [2] C. Di Blasi, Polym. Int. 49 (2000) 1133–1146.
- [3] M.G. Gronli, PhD thesis, NTNU, Trondheim, 1996.
- [4] J. Larfeldt, PhD thesis, Chalmers Tekniska Hogskola, Goteborg, 2000.
- [5] J. Larfeldt, B. Leckner, M.C. Melaaen, Fuel 70 (2000) 1637–1643.
- [6] M.G. Gronli, M.C. Melaaen, Energy Fuels 14 (2000) 791–800.
- [7] A.M.C. Janse, R.W.J. Westerhout, W. Prins, Chem. Eng. Proc. 39 (2000) 239–252.
- [8] C. Di Blasi, Chem. Eng. Sci. 55 (2000) 5999–6013.
- [9] J. Larfeldt, B. Leckner, M.C. Melaaen, in: A.V. Bridgewater (Ed.), Progress in Thermochemical Biomass Conversion, Blackwell Science, Oxford, 2001, pp. 1046–1060.
- [10] C. Di Blasi, AIChE J. 48 (2002) 2386–2397.
- [11] K.M. Bryden, K.W. Ragland, C.J. Rutland, Biomass Bioenergy 22 (2002) 41–53.
- [12] M.J. Hagge, K.M. Bryden, Chem. Eng. Sci. 57 (2002) 2811–2823.
- [13] M. Bellais, K.O. Davidsson, T. Liliendahl, K. Sjostrom, J.B.C. Pettersson, Fuel 82 (2003) 1541–1548.
- [14] K.M. Bryden, M.J. Hagge, Fuel 82 (2003) 1633–1644.
- [15] B.V. Babu, A.S. Chaurasia, Chem. Eng. Sci. 59 (2004) 611–622.
- [16] B.V. Babu, A.S. Chaurasia, Chem. Eng. Sci. 59 (2004) 1999–2012.
- [17] P.S. Maa, R.C. Bailie, Combust. Sci. Technol. 7 (1973) 257–269.
- [18] Y. Chen, M.A. Delichatios, V. Motevalli, Combust. Sci. Technol. 88 (1993) 309–328.
- [19] B. Moghtaderi, V. Novozhilov, D. Fletcher, J.H. Kent, Fire Mater. 21 (1997) 7–16.
- [20] F. Jia, E.R. Galea, M.K. Patel, Fire Mater. 23 (1999) 71–78.
- [21] M.J. Spearpoint, J.G. Quintiere, Combust. Flame 123 (2000) 308–324.
- [22] M.J. Spearpoint, J.G. Quintiere, Fire Safety J. 36 (2001) 391–415.
- [23] A. Galgano, C. Di Blasi, Ind. Eng. Chem. Res. 42 (2003) 2101–2111.
- [24] G.I. Palchonok, V.A. Dikalenko, L.K. Stanchits, V.A. Borodulya, J. Werther, B. Leckner, in: F.D.S. Preto (Ed.), Proceedings of the 14th International Conference on Fluidized Bed Combustion, Am. Soc. Mech. Eng., New York, 1997, pp. 125–133.
- [25] D.L. Simms, M. Law, Combust. Flame 11 (1967) 377–388.
- [26] C.K. Lee, J.R. Diehl, Combust. Flame 42 (1981) 123–138.
- [27] M.R. Gray, W.H. Corcoran, G.R. Gavalas, Ind. Eng. Chem. Res. 24 (1985) 646–651.
- [28] W.R. Chan, M. Kelbon, B. Krieger-Brockett, Ind. Eng. Chem. Res. 27 (1988) 2261–2275.
- [29] C. Di Blasi, E. Gonzalez Hernandez, A. Santoro, Ind. Eng. Chem. Res. 39 (2000) 873–882.
- [30] R.H. White, E.L. Schaffer, Wood Fiber 13 (1981) 17–38.
- [31] B. Fredlund, Fire Safety J. 20 (1993) 39–69.
- [32] A.M. Kanury, P.L. Blackshear, Combust. Sci. Technol. 1 (1970) 339–355.
- [33] C. Di Blasi, C. Branca, A. Santoro, E. Gonzalez Hernandez, Combust. Flame 124 (2001) 165–177.
- [34] R.B. Bird, W.E. Stewart, E.N. Lightfoot, Transport Phenomena, second ed., Wiley, New York, 2002, p. 419.
- [35] C. Di Blasi, Chem. Eng. Sci. 51 (1996) 1121–1132.
- [36] S.S. Alves, J.L. Figueiredo, Chem. Eng. Sci. 44 (1989) 2861–2869.
- [37] C. Di Blasi, Chem. Eng. Sci. 53 (1998) 353–366.
- [38] C. Di Blasi, C. Branca, S. Sparano, B. La Mantia, Biomass Bioenergy 25 (2003) 45–58.
- [39] K.M. Bryden, K.W. Ragland, in: A.V. Bridgewater, D. Boocock (Eds.), Developments in Thermochemical Biomass Conversion, Blackie, 1997, pp. 1331–1345.
- [40] R.H. Perry, D. Green, Perry's Chemical Engineers' Handbook, McGraw-Hill, New York, 1985.
- [41] C. Di Blasi, C. Branca, Ind. Eng. Chem. Res. 40 (2001) 5547–5556.